RiskQuant · Credit VaR

Credit losses when counterparties default together

2026-11-06

EXECUTIVE SUMMARY

Part 0 · Current Status

What Credit VaR needs, what exists today, and what can run now

> ***The default models and the Excel tool* exist today. Exposure at default is the one input that needs more work.***

** The Excel tool is effectively deprecated. The web app replaces it, with delivery planned for November 2026.*

What Credit VaR Needs

Credit VaR needs four inputs. Exposure at default (EAD) and loss given default (LGD) set the size of each loss. Probability of default (PD) and default dependence set how often counterparties default, alone or together.

#	Input	In plain terms
1	**Exposure at default (EAD)**	The amount we are owed when a counterparty defaults
2	**Loss given default (LGD)**	The share of that amount we lose after recovery
3	**Probability of default (PD)**	The chance that each counterparty defaults
4	**Default dependence**	The chance that counterparties default together

> ***EAD, LGD and PD set the expected loss in most models. If a model keeps each counterparty's PD, default dependence leaves the expected loss unchanged and changes the tail. Some models replace or scale the PD, so they also change the expected loss.***

What We Already Have

LGD, PD and default dependence exist today. Exposure at default exists in part.

#	Input	Status
1	EAD	**Partial** · The current EAD input is stressed Replacement Cost (RC): net MTM per netting set after any collateral, floored at zero. SA-CCR EAD is separate: 1.4 × (RC + PFE), where PFE (potential future exposure) is a supervisory add-on. SA-CCR covers unmargined linear equity and commodity trades only.
2	LGD	**Have** · input per counterparty
3	PD	**Have** · input per counterparty
4	Default dependence	**Have** · the `riskquant` model suite is built

> ***Credit VaR runs today with the (stressed) Replacement Cost as EAD. A forward-looking exposure model, which gives exposure over time, is planned as a separate investment.***

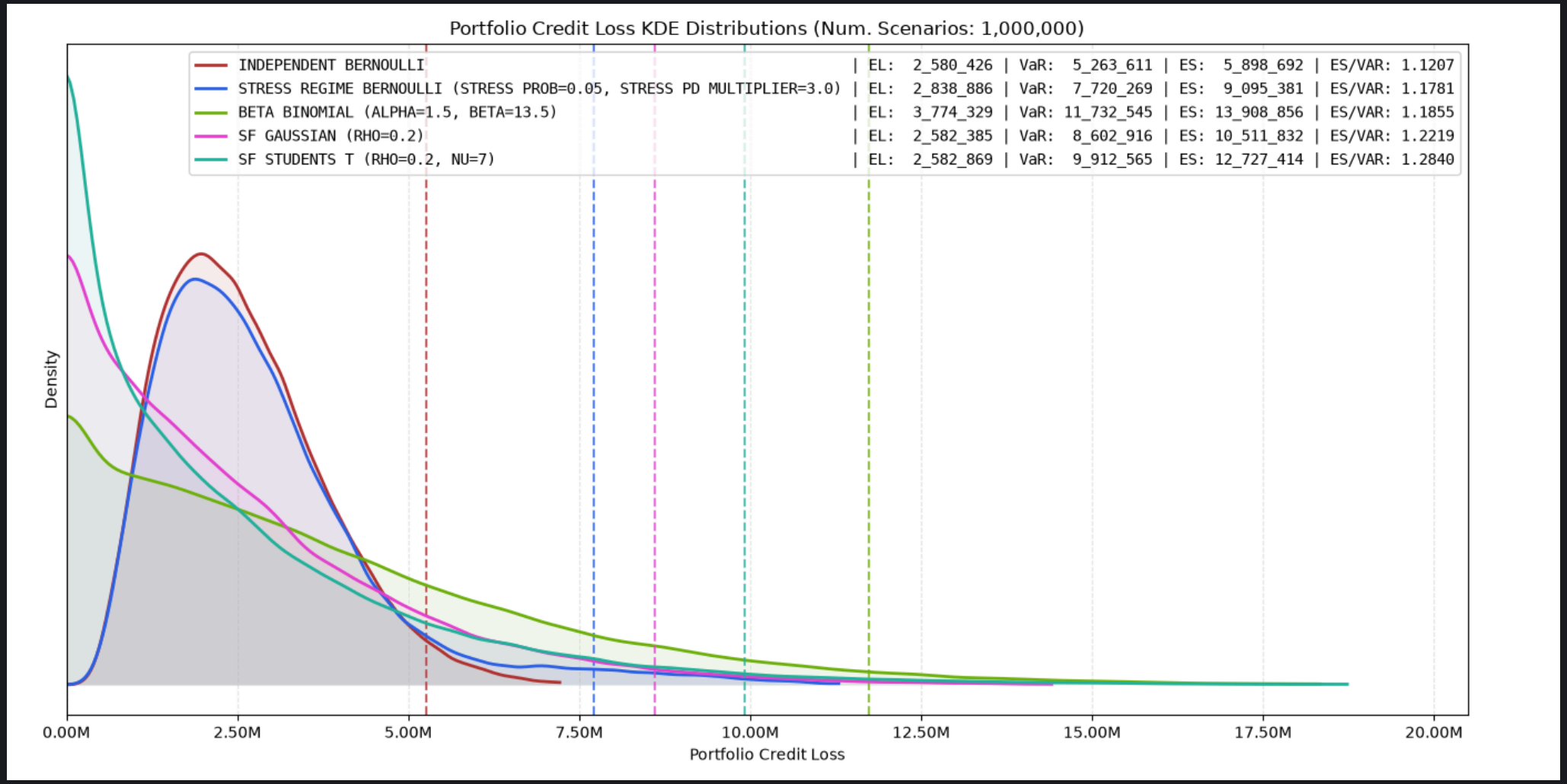
What We Can Do Today

Each model runs on the current or stressed Replacement Cost (RC) as EAD. In each scenario, the loss is EAD × LGD, summed over the counterparties that default.

Model	What it assumes
Independent Bernoulli	Each counterparty defaults on its own. No default clustering.
Stress-Regime Bernoulli	In some scenarios a shared stress state multiplies every PD.
Beta-Binomial	One shared default probability is drawn for each scenario.
Single-Factor Gaussian Copula	One shared factor drives all defaults.
Single-Factor Student-t Copula	The same shared factor, with more default clustering in extreme scenarios.

> ***All five models use the same portfolio. Each model allows a different amount of default clustering, so the tail loss can differ. A deterministic upper bound (Forced Default) is also available, as are other models.***

STX B.V. - GOO - Base Case



Credit VaR is Operational Today

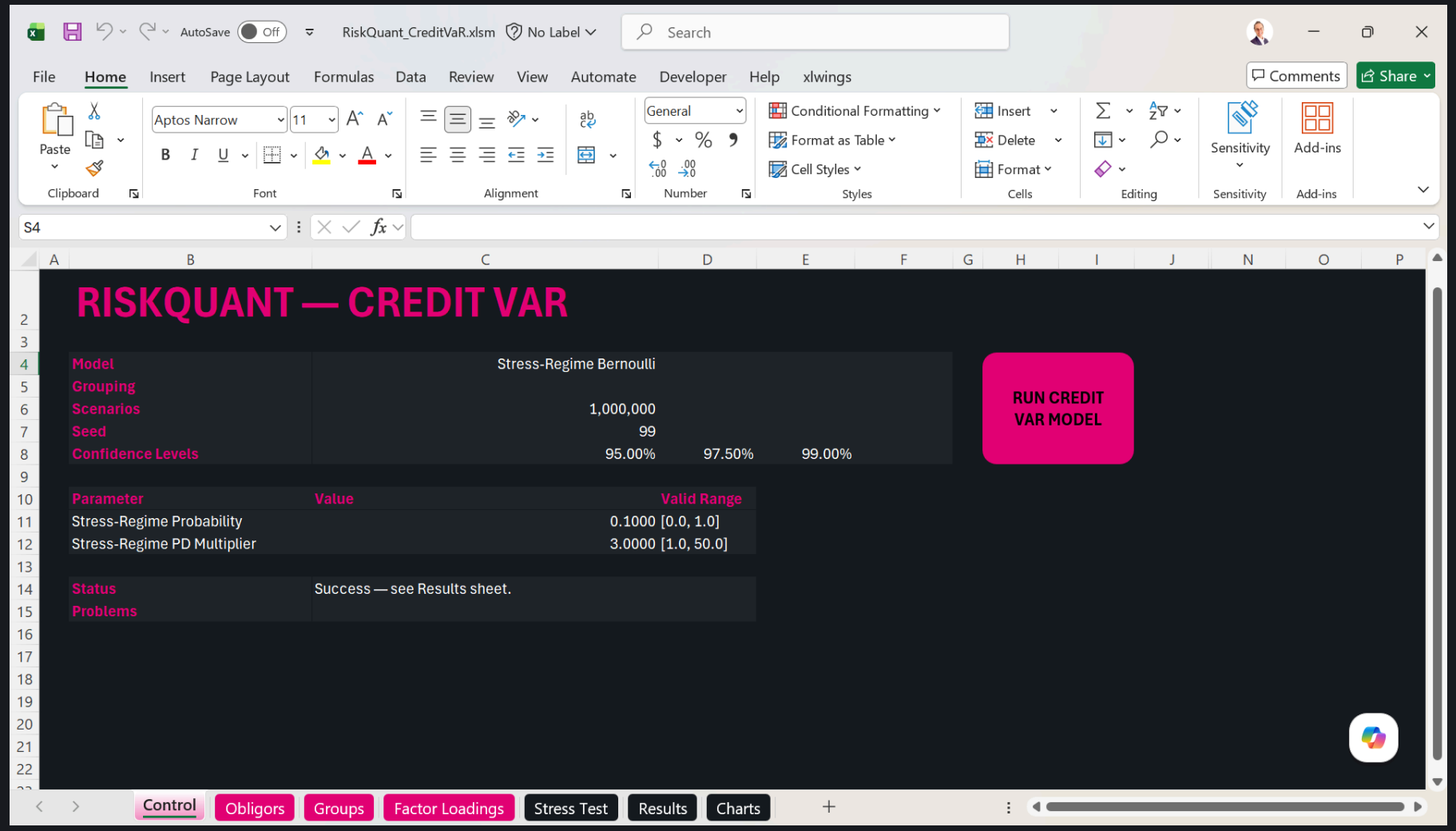
The Excel tool runs Credit VaR on an uploaded portfolio.

- **Configure:** Enter counterparties with EAD, PD and LGD. Select a model.
- **Run:** One click returns the loss distribution, VaR, ES and unexpected credit loss (UL).
- **Attribute:** ES contribution and incremental VaR/ES show which counterparties drive the tail.
- **Stress:** Force chosen counterparties to default in the simulation, or list them for a what-if loss. Change exposures outside the tool and re-run.
- **Exposure:** EAD is the (stressed) Replacement Cost. A forward-looking model is next.

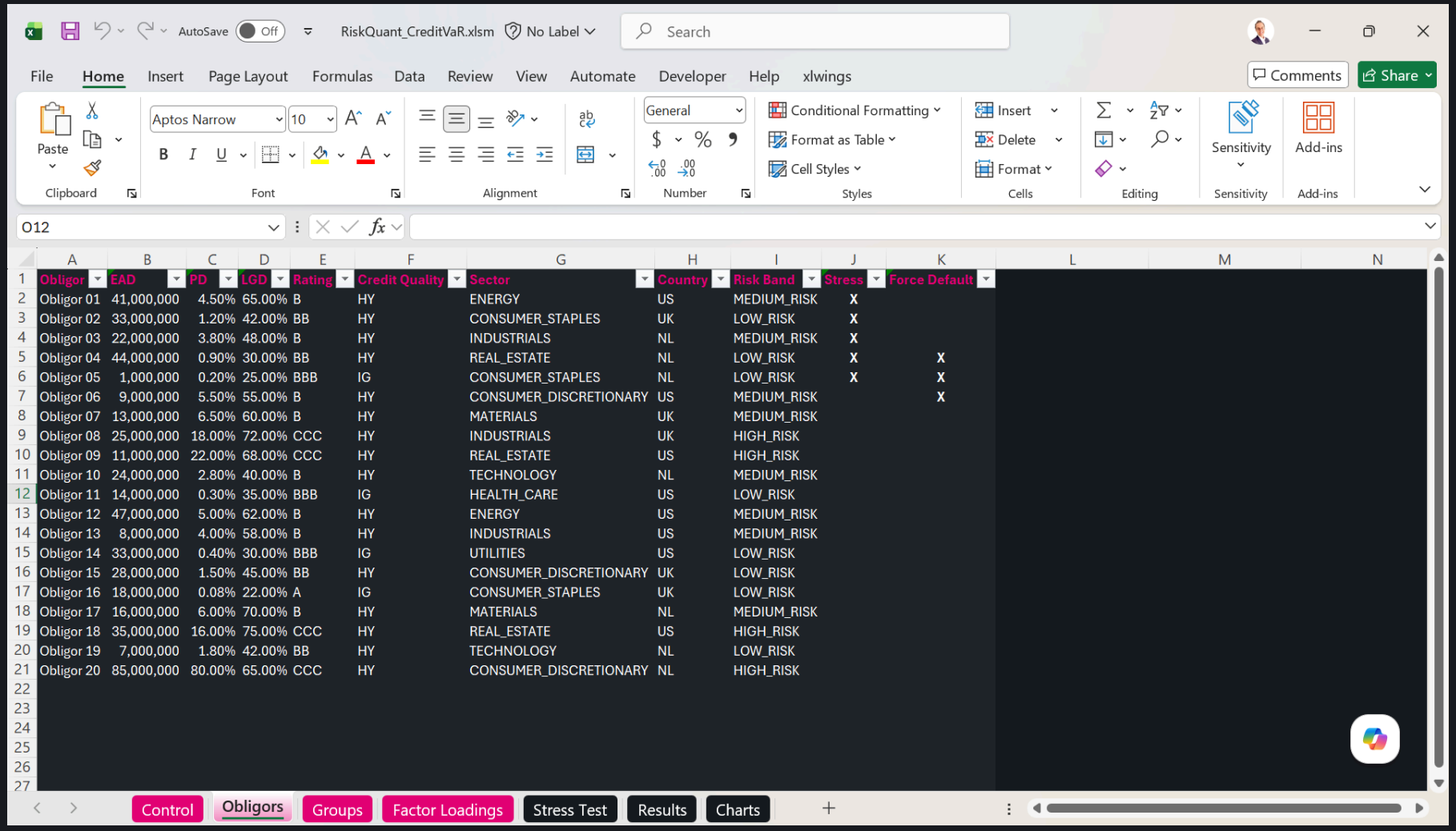
> ***Each run records the model, the inputs, the seed and the software version.***

> ***The Excel tool is deprecated in favour of a web app (planned Nov-2026).***

RiskQuant Credit VaR Excel Tool



RiskQuant Credit VaR Excel Tool



Obligor	EAD	PD	LGD	Rating	Credit Quality	Sector	Country	Risk Band	Stress	Force Default
Obligor 01	41,000,000	4.50%	65.00%	B	HY	ENERGY	US	MEDIUM_RISK	X	
Obligor 02	33,000,000	1.20%	42.00%	BB	HY	CONSUMER_STAPLES	UK	LOW_RISK	X	
Obligor 03	22,000,000	3.80%	48.00%	B	HY	INDUSTRIALS	NL	MEDIUM_RISK	X	
Obligor 04	44,000,000	0.90%	30.00%	BB	HY	REAL_ESTATE	NL	LOW_RISK	X	X
Obligor 05	1,000,000	0.20%	25.00%	BBB	IG	CONSUMER_STAPLES	NL	LOW_RISK	X	X
Obligor 06	9,000,000	5.50%	55.00%	B	HY	CONSUMER_DISCRETIONARY	US	MEDIUM_RISK		X
Obligor 07	13,000,000	6.50%	60.00%	B	HY	MATERIALS	UK	MEDIUM_RISK		
Obligor 08	25,000,000	18.00%	72.00%	CCC	HY	INDUSTRIALS	UK	HIGH_RISK		
Obligor 09	11,000,000	22.00%	68.00%	CCC	HY	REAL_ESTATE	US	HIGH_RISK		
Obligor 10	24,000,000	2.80%	40.00%	B	HY	TECHNOLOGY	NL	MEDIUM_RISK		
Obligor 11	14,000,000	0.30%	35.00%	BBB	IG	HEALTH_CARE	US	LOW_RISK		
Obligor 12	47,000,000	5.00%	62.00%	B	HY	ENERGY	US	MEDIUM_RISK		
Obligor 13	8,000,000	4.00%	58.00%	B	HY	INDUSTRIALS	US	MEDIUM_RISK		
Obligor 14	33,000,000	0.40%	30.00%	BBB	IG	UTILITIES	US	LOW_RISK		
Obligor 15	28,000,000	1.50%	45.00%	BB	HY	CONSUMER_DISCRETIONARY	UK	LOW_RISK		
Obligor 16	18,000,000	0.08%	22.00%	A	IG	CONSUMER_STAPLES	UK	LOW_RISK		
Obligor 17	16,000,000	6.00%	70.00%	B	HY	MATERIALS	NL	MEDIUM_RISK		
Obligor 18	35,000,000	16.00%	75.00%	CCC	HY	REAL_ESTATE	US	HIGH_RISK		
Obligor 19	7,000,000	1.80%	42.00%	BB	HY	TECHNOLOGY	NL	LOW_RISK		
Obligor 20	85,000,000	80.00%	65.00%	CCC	HY	CONSUMER_DISCRETIONARY	NL	HIGH_RISK		

FORBIDDEN MATERIALS

Part 1 · Credit VaR

Default dependence assumptions change the tail loss

What Credit VaR Measures

Credit VaR estimates the losses that our trading portfolio could suffer if counterparties default.

Component	Question	Determines
Exposure and LGD	How much do we lose if a counterparty defaults?	The size of each loss
Default probability	How likely is each counterparty to default?	The expected loss
Default dependence	How likely are counterparties to **default together**?	**How strongly extreme losses cluster**

Expected Loss and Credit VaR

Expected Loss is the average counterparty credit loss.

Credit VaR is the credit loss that scenarios exceed with a small, chosen probability (e.g., 1%, which is a 99% confidence level).

Expected Loss	Credit VaR
"What is our expected credit loss?"	"What loss could we face in a severe event?"
Driven by exposure, LGD and PD	Driven by exposure, LGD, PD and **default dependence**
Typically used to estimate the expected cost of counterparty default	Typically used to inform capital and risk appetite

> ***Each model sets how strongly counterparties default together. Credit VaR includes that effect in the tail.***

Questions Credit VaR Helps Answer

- **How much** could we lose in a severe counterparty default event?
- **Which counterparties** contribute most to the tail loss?
- **How much** would the tail fall if we exit one counterparty?
- **What is the loss** if a chosen group of counterparties defaults together?
- **How much** does the tail change when the dependence assumption changes?

> ***Each model assumes a different amount of default clustering.***

Inputs to Credit VaR

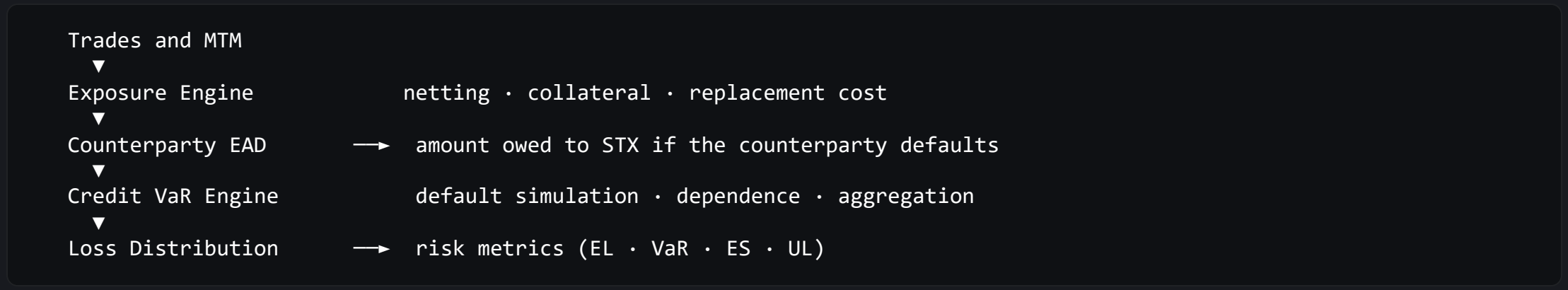
Credit VaR uses **four** inputs. EAD and LGD set the **size of a loss**. PD and default dependence set **how often losses occur and how much they cluster**.

Input	Question answered	Source
Exposure at default (EAD)	How much value is exposed if a counterparty defaults?	Exposure engine (RC) or an external model
Loss given default (LGD)	How much of that exposure is lost after recovery?	Counterparty table
Probability of default (PD)	How likely is each counterparty to default?	Counterparty table
Default dependence	How likely are defaults to occur together?	Model choice and parameters (ρ, ν)

> ***The loss is EAD × LGD when a counterparty defaults. Default dependence links the defaults of counterparties in the same scenario.***

Where Credit VaR Fits

- The **exposure engine** calculates EAD. Today the EAD supplied to Credit VaR is the current or stressed Replacement Cost (RC). A forward-looking model (exposure over time) is not available yet.
- The **Credit VaR engine** simulates defaults and sums the losses.
- A credit loss arises only when a counterparty's trades have positive net replacement value at default.



Part 2 · Jump-to-Default (JTD)

From counterparty defaults to portfolio loss distributions

> ***In each scenario, default dependence sets how many counterparties default together.***

Jump-to-Default: Summary

The event: a counterparty defaults. Our positive net exposure to it is the **exposure at default (EAD)**. The credit loss is **EAD × LGD**, where **LGD** is the share of **EAD** that we do not recover.

Exposure is the net mark-to-market that a counterparty owes us, after any collateral. Trades offset only inside the same legally enforceable netting set. Exposures of different netting sets add up. Netting sets are not yet clearly defined.

Each scenario: take the EAD of each counterparty, decide which counterparties default, apply the LGD, and sum the losses. Repeat for 250,000 scenarios (default). The result is the portfolio loss distribution.

> ***The model must state how likely another counterparty is to default in the same scenario as the first.***

What JTD Credit VaR Does Not Answer

JTD Credit VaR measures the loss **when default happens**.

It does not answer:

Question	Where it sits
How large could exposure become before default?	CCR modelling
What is the loss from a sudden move in market prices?	Market risk
Does exposure increase as the counterparty's credit quality falls (wrong-way risk)?	Not modelled. The model does not capture wrong-way risk. In the simulation, EAD is a fixed input.
How long does close-out take after default?	CCR modelling
How does credit quality fall before default?	Not modelled. This needs a credit migration model.

> ***JTD Credit VaR measures default loss only. It is one part of total counterparty risk.***

What Drives Expected Loss and Tail Loss

Driven by exposure, LGD and PD	Affected by default dependence
Expected Loss	**The tail: VaR and ES**
Same for every model that keeps each counterparty's PD	Can differ strongly between models

Example: €100m exposure, 40% recovery → **€60m loss if it defaults.**

Models that keep the counterparty's PD give it the same expected loss. They differ on how often it defaults in the same scenario as twenty others.

Five Models on the Same Portfolio

The common question: **when one counterparty defaults, how likely are others to default too?**

#	Model	Assumption added
0	Forced Default (bound)	Every counterparty defaults. No probability or dependence.
1	Independent Bernoulli	Defaults are unrelated
2	Beta-Binomial	Default rates can move higher or lower together
3	Stress-Regime Bernoulli	The market can enter a crisis state
4	Gaussian Copula	Counterparties share common risk drivers
5	Student-t Copula	Extreme tail dependence

> ***Independent Bernoulli and the copulas keep each counterparty's PD, so their expected loss is the same (apart from simulation error). Beta-Binomial and Stress-Regime change the PD.***

0 · Forced Default (bound): *all counterparties default*

Intuition: every counterparty defaults in every scenario. PD plays no part. This is a deterministic bound. It has no dependence assumption.

Answers: *what is the largest default loss for the given EAD and LGD?*

Strengths	Weaknesses
Needs only EAD and LGD	Not a realistic scenario
Gives an upper bound for the given EAD and LGD	Not suitable as a capital estimate

Use it for: an upper-bound check, new portfolios with no PD history, and hard stress cases.

> ***For the same EAD and LGD, no model gives a loss above this value. A change in EAD or LGD changes the bound.***

1 · Independent Bernoulli: *no default clustering*

Intuition: each counterparty defaults with its own PD. A default does not change the chance of any other default.

Answers: *what does the loss look like if defaults are unrelated?*

Strengths	Weaknesses
Simple, fast, gives a full distribution	Does not allow default clustering
Captures single-counterparty concentration	**Understates the tail and overstates diversification**

Use it for: granular, homogeneous (sub)portfolios, and as the baseline for the other models.

> ***Counterparties share common drivers, so this model overstates diversification.***

2 · Beta-Binomial: *"one shared default rate"*

Intuition: the model draws one default probability for the whole portfolio in each scenario. Every counterparty uses that probability. In a high-rate scenario, many counterparties default together.

Answers: *what happens when the portfolio default rate changes from scenario to scenario?*

Strengths	Weaknesses
Creates default clustering without specifying risk factors	One pool: no sector, geography or counterparty drivers
Introduces volatility in the portfolio default rate	Ignores each counterparty's own PD

Use it for: portfolios where default uncertainty matters and a factor model is not required.

> ***The Beta parameters set the average PD for every counterparty.***

3 · Stress-Regime Bernoulli: *a shared stress state*

Intuition: in some scenarios a shared stress state is active. In those scenarios the model multiplies every PD by a fixed factor, up to 100%.

Answers: *what happens when a defined stress state raises all PDs together?*

Strengths	Weaknesses
Makes default dependence explicit and interpretable	Assumes only two states: normal and stress
Easy to link losses to defined stress scenarios	Requires judgement on stress severity and likelihood

Use it for: stress testing, limit discussions and adverse-scenario analysis.

> ***One shared stress state links all counterparties. The state also raises expected loss.***

4 · Gaussian Copula: *one shared factor*

Intuition: each counterparty has its own component and a shared component. When the shared factor is low, many counterparties weaken together and defaults cluster.

Answers: *how much do shared drivers reduce our diversification?*

Strengths	Weaknesses
Standard market model (Vasicek, Basel ASRF)	One shared factor only
One correlation parameter (ρ) controls clustering	Gaussian tails understate joint extreme defaults

Use it for: portfolio tail risk measurement and counterparty risk capital.

> ***The Student-t copula adds tail dependence to this model.***

5 · Student-t Copula: *stronger tail dependence*

Intuition: the same shared factor as the Gaussian copula, plus a shared random scale. In extreme scenarios, counterparties are more likely to default together than under the Gaussian copula.

Answers: *how large can the loss become when defaults cluster in the tail?*

Strengths	Weaknesses
Captures stronger tail dependence	One extra parameter (degrees of freedom, ν) needs a value
Allows defaults to cluster in extreme scenarios	Tail loss can exceed the Gaussian copula result at the same ρ, depending on the portfolio and the confidence level.

Use it for: extreme portfolio losses and stress scenarios where diversification falls.

> ***In extreme scenarios, diversification between counterparties falls.***

Comparing the Models

Model	Realism	Transparency	Tail capture	Data need	Business fit
Forced Default (bound)	n/a	High	n/a	Minimal	Stress bound
Independent Bernoulli	Low	High	Low	Low	Baseline view
Beta-Binomial	Med	High	Med	Low	Portfolio stress
Stress-Regime	Med	High	Med-High	Med	Scenario analysis
Gaussian Copula	High	Med	Med	Med-High	Core risk view
Student-t Copula	**High**	Med	**High**	Med-High	Extreme tail view

> ***Simple models are easier to explain. Richer models capture more dependence. No single model is best in every column, so we run several.***

Which Model Answers Which Question?

The question we need to answer	Model
What is the largest default loss for the given EAD and LGD?	Forced Default (bound)
What if defaults are independent?	Independent Bernoulli
What if the portfolio default rate is higher or lower than expected?	Beta-Binomial
What happens in a defined stress scenario?	Stress-Regime Bernoulli
What if counterparties are exposed to the same economic drivers?	Gaussian Copula
What if extreme events create unusually strong clustering?	Student-t Copula

> ***Compare the models to see how much the dependence assumption changes the tail.***

Part 3 · Reading the Output

Metrics and attribution

> ***Credit VaR also shows which counterparties drive the tail.***

The Metrics Every Model Returns

Metric	Reads as	Feeds
Expected Loss (EL)	average counterparty default loss	expected credit cost
Loss volatility (σ)	how much the portfolio loss varies	unexpected loss
VaR	the loss exceeded with probability 1 − α (α = confidence level)	risk-appetite limits
Expected Shortfall (ES)	the *average* loss in the scenarios beyond VaR	internal capital
Unexpected Credit Loss (UL)	alternative measures: VaR − EL (VaR-based) and ES − EL (ES-based)	internal capital

> ***VaR is one point of the loss distribution. ES is the average of all losses beyond that point.***

Models that keep each counterparty's PD give the same EL, apart from simulation error. VaR and ES can differ between models.

Incremental VaR & ES: *what each counterparty adds*

Hold the portfolio constant, remove one counterparty and re-measure the tail. The difference is its incremental risk.

```
incremental_risk(i) = risk(portfolio) − risk(portfolio excl. i)
```

Answers: *how much would the tail fall if we exit this counterparty?*

Property	Consequence
Computed for both VaR and ES	Same simulated losses, so no new simulation run
Not additive	Incremental values do not sum to portfolio risk

Use it for: limit discussions, novation and unwind decisions for existing counterparties. A new counterparty or trade needs a new run with the changed EAD.

Contribution ES

Contribution ES splits today's ES across the counterparties. The contributions add up exactly to the portfolio ES.

> `contribution_ES(i)` = ***average loss of counterparty*** `i` ***in the portfolio's worst*** `1−α` ***share of scenarios.***

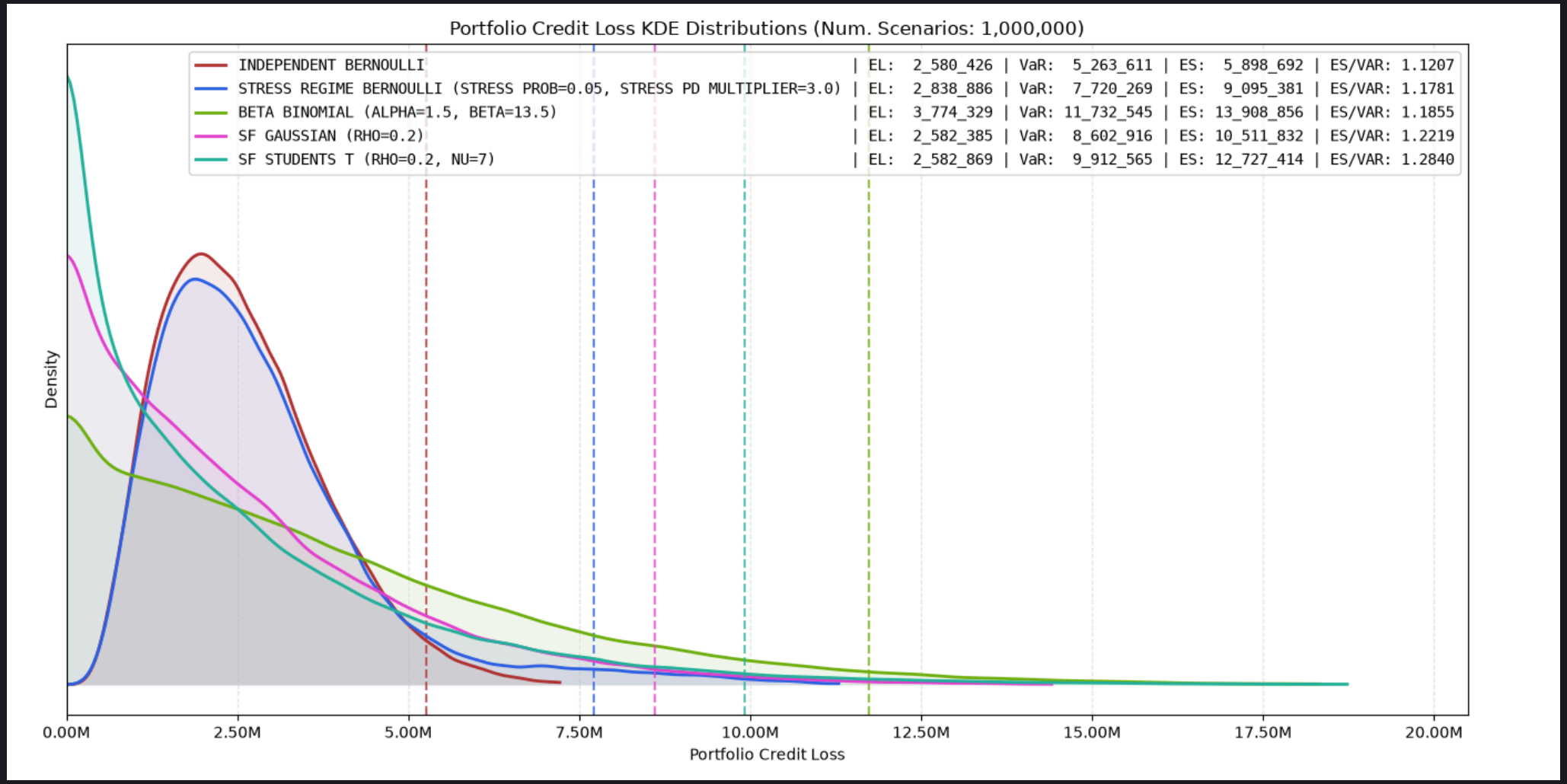
> `portfolio_ES = sum_i contribution_ES(i)`

RiskQuant · Credit VaR

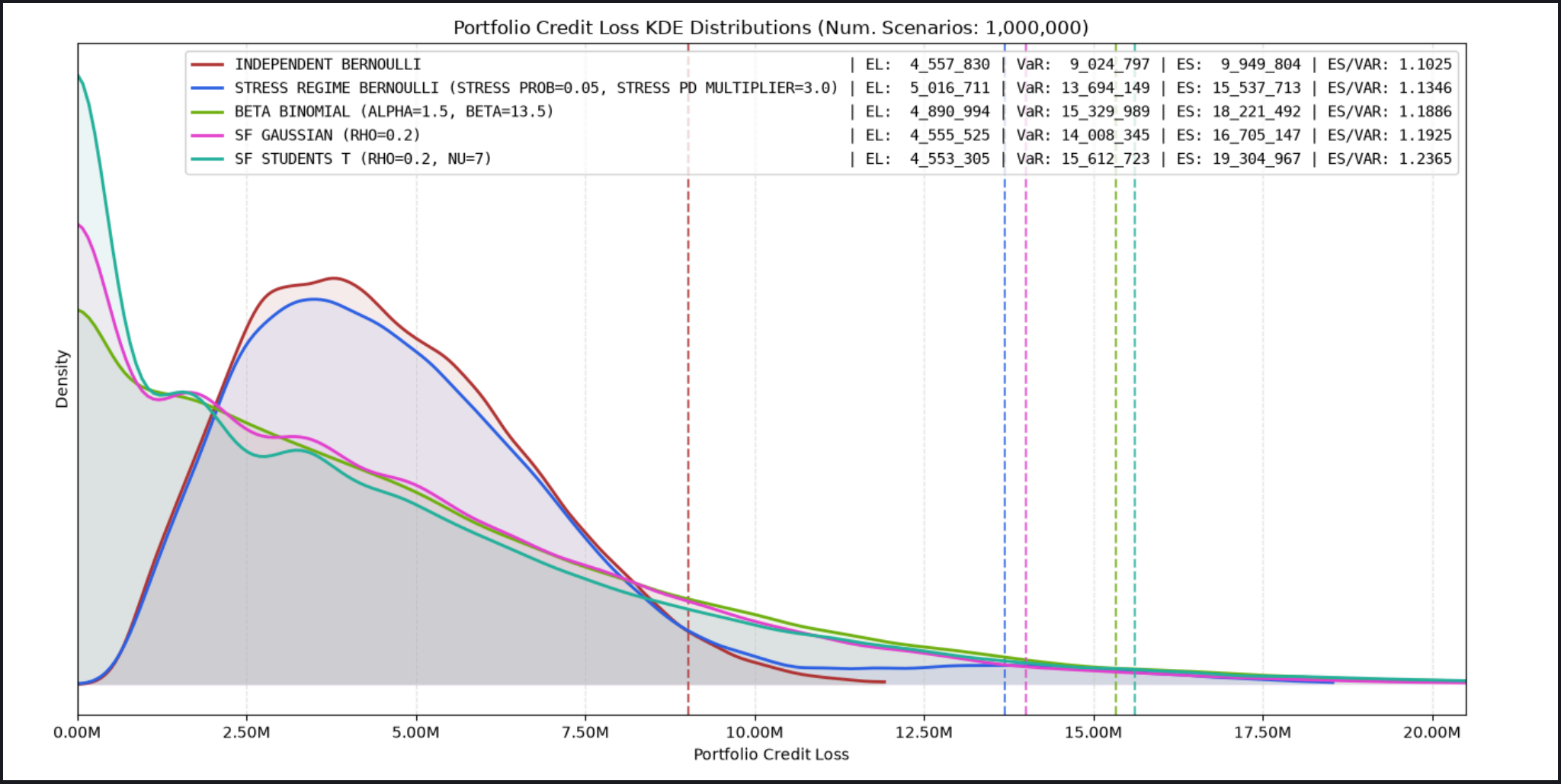
> ***Credit VaR measures the tail of the credit loss distribution. Default dependence changes that tail and can make it larger.***

Appendix: STX Credit VaR

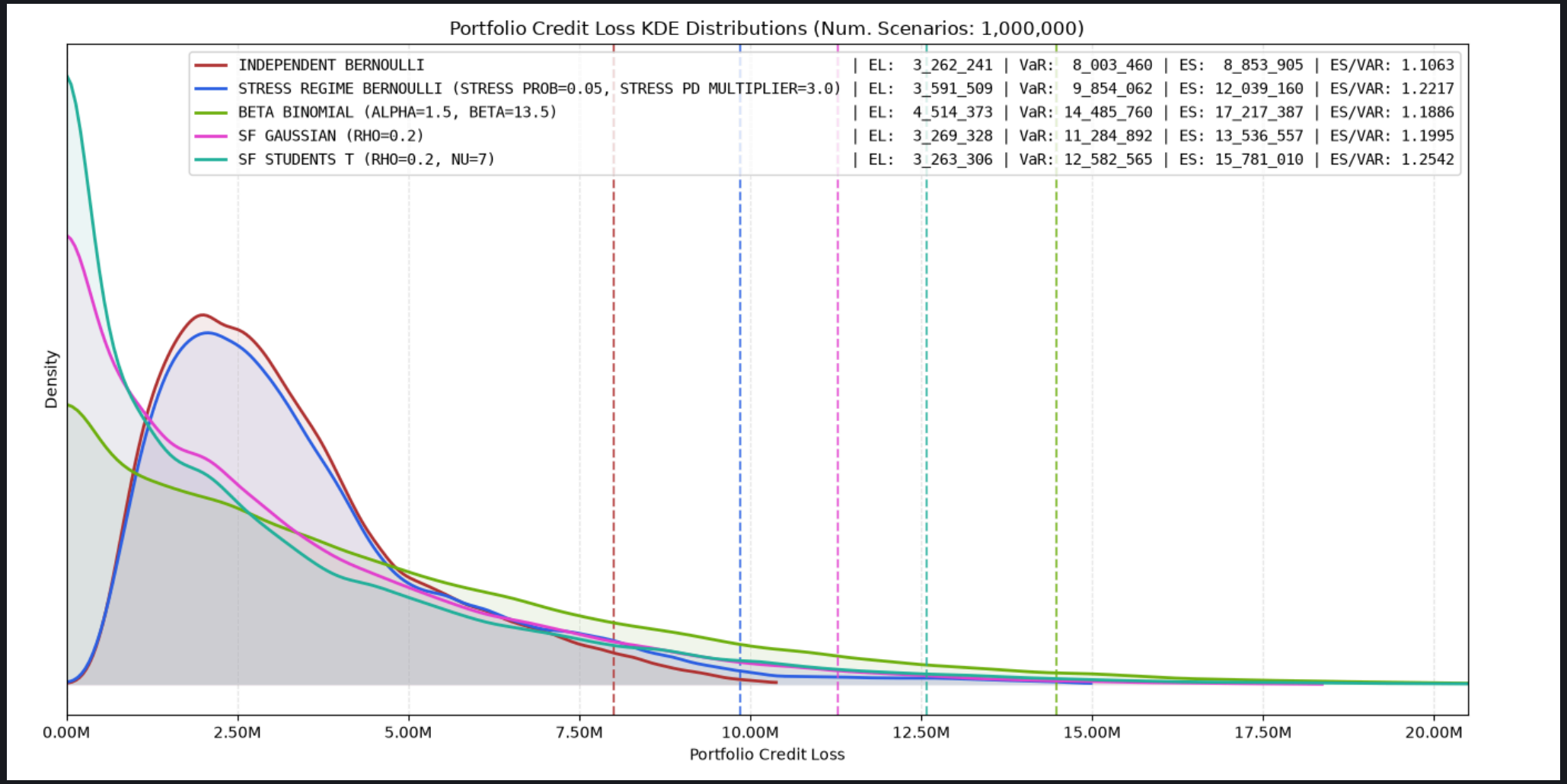
STX B.V. - GOO - Base Case



STX B.V. - GOO - Stressed (+20%)



STX B.V. - GOO - Stressed (-20%)



STX B.V. - GOO - Random Up/Down 10%-20% (PC Level)

